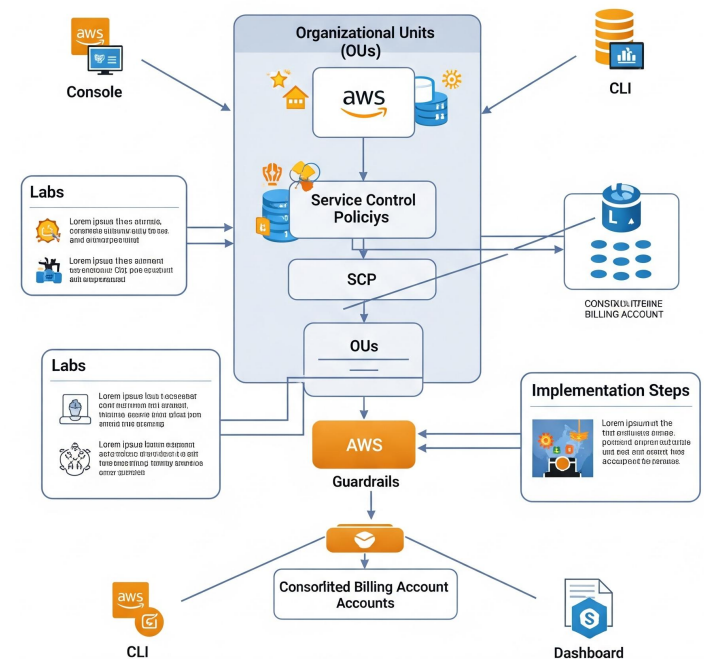


AWS Organizations & Control Tower

- Topic: AWS Organizations & Control Tower
- Covers SCPs, OUs, Billing, Guardrails
- Includes AWS Console, CLI, Dashboard
- Focus on governance & automation
- Designed for enterprise-scale AWS
- Includes labs & implementation steps

AWS Organizations and Control Tower

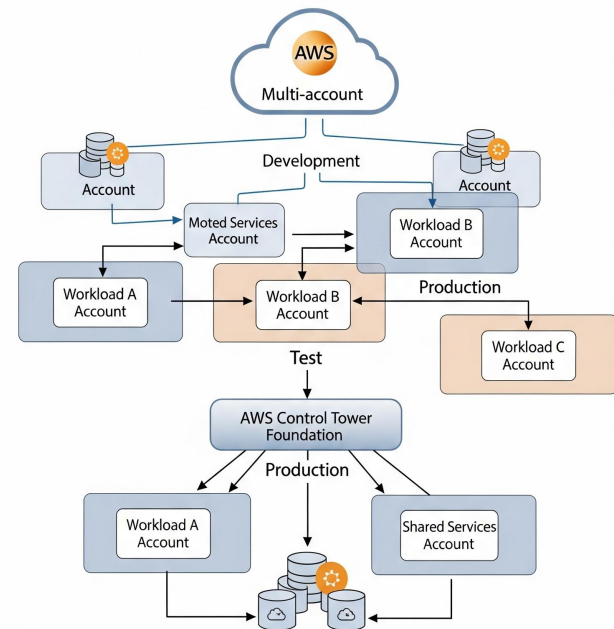


Learning Objectives

- Understand AWS multi-account management
- Explore Service Control Policies (SCPs)
- Learn consolidated billing structure
- Configure Control Tower landing zones
- Automate account vending & guardrails
- Validate real-world enterprise setup

Why Multi-Account Architecture?

- Security isolation between workloads
- Clear billing and accountability
- Minimize blast radius of breaches
- Environment separation (Dev/Test/Prod)
- Easier compliance management
- Foundation for Control Tower



Introduction to AWS Organizations

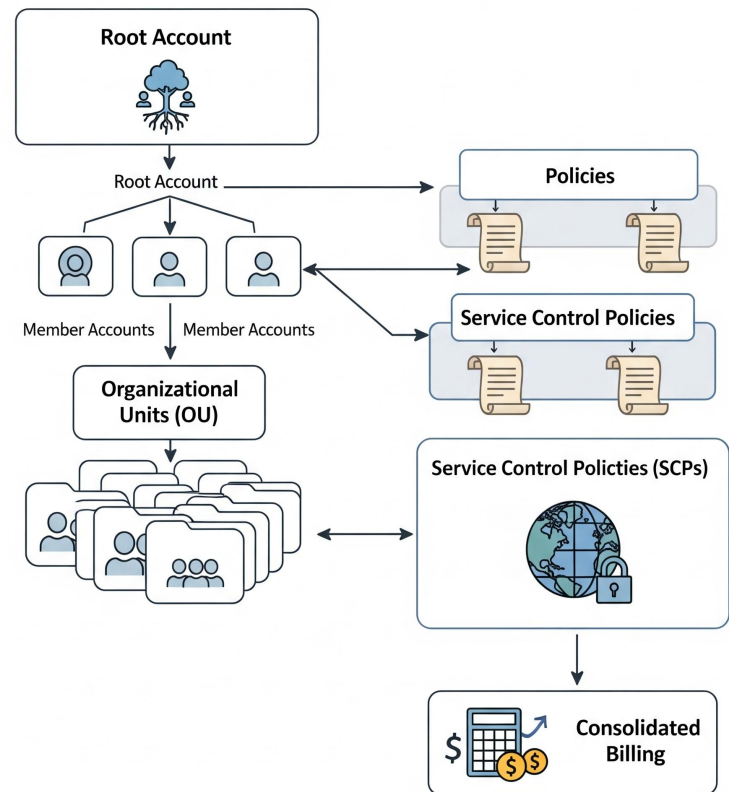
- Centralized account management tool
- Enables unified governance
- Allows automated policy enforcement
- Facilitates consolidated billing
- Integrates with Control Tower
- Provides scalable OU structure

AWS Organizations



Components of AWS Organizations

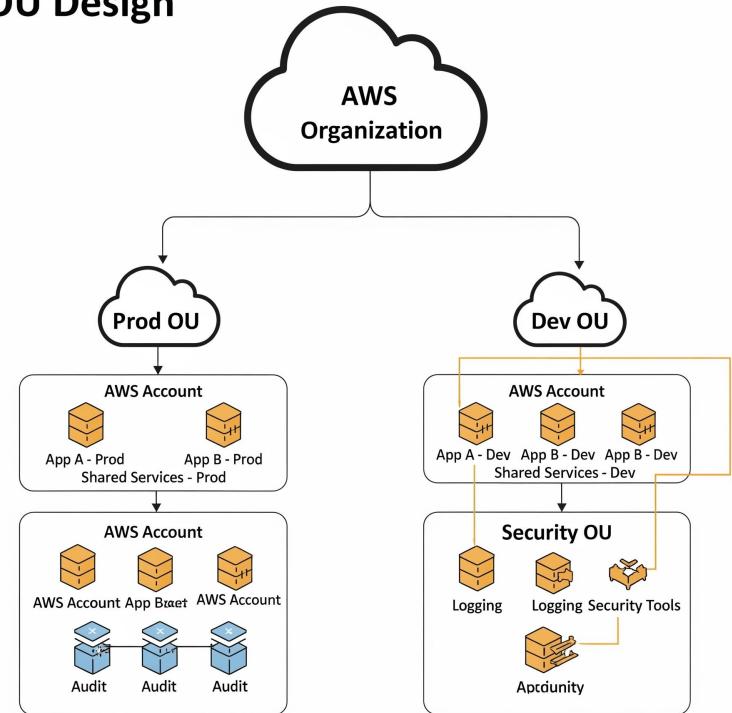
- Root account – top-level control
- Member accounts – linked accounts
- OUs – group of related accounts
- Policies – enforce governance rules
- SCPs – control permissions globally
- Consolidated Billing – unified cost view



Organizational Unit (OU) Design

- Logical grouping of AWS accounts
- Example: Prod, Dev, Security OUs
- Hierarchical structure with inheritance
- Centralized policy enforcement
- Granular access control via SCPs
- Enables tailored governance

OU Design



Service Control Policies (SCPs)

- Define account-level permission boundaries
- Apply to OUs or specific accounts
- JSON-based policy format
- Complement IAM policies (not replace)
- Control what users can't do
- Essential for security compliance



SCP Example

- Policy JSON with version and statements
- Denies EC2 termination globally
- Demonstrates fine-grained control
- Enforced across multiple accounts
- Propagates down OU hierarchy
- Example: Deny ec2:TerminateInstances

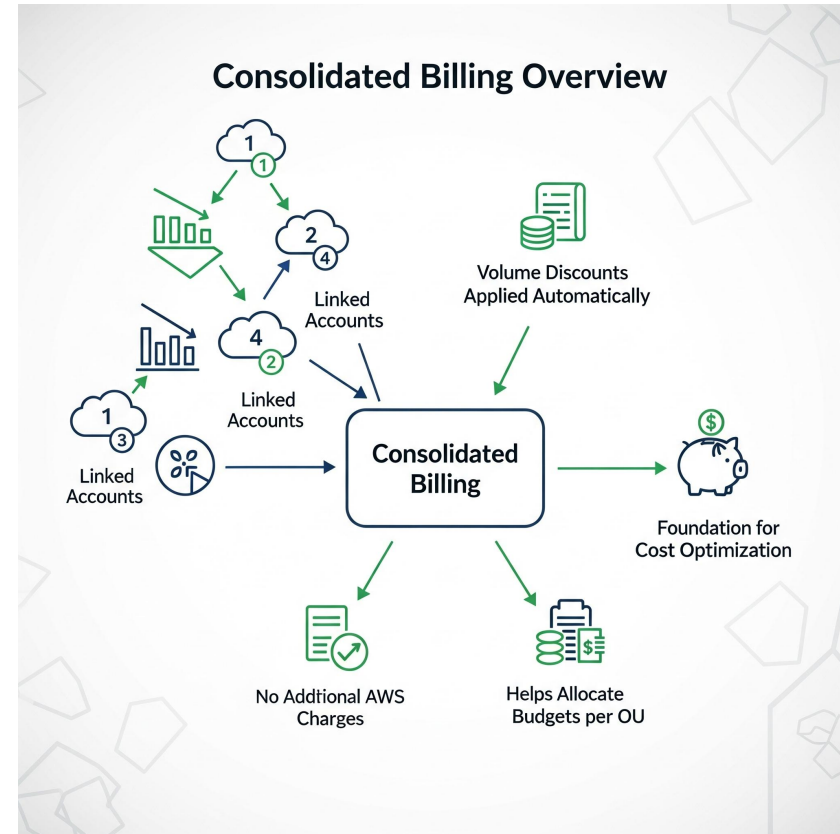


SCP Best Practices

- Start with FullAWSAccess baseline
- Add Deny policies incrementally
- Test in non-prod first
- Review policies regularly
- Avoid Deny on critical AWS services
- Monitor with AWS CloudTrail

Consolidated Billing Overview

- Combines charges from all linked accounts
- Volume discounts applied automatically
- Centralized invoice for simplicity
- No additional AWS charges
- Helps allocate budgets per OU
- Foundation for cost optimization



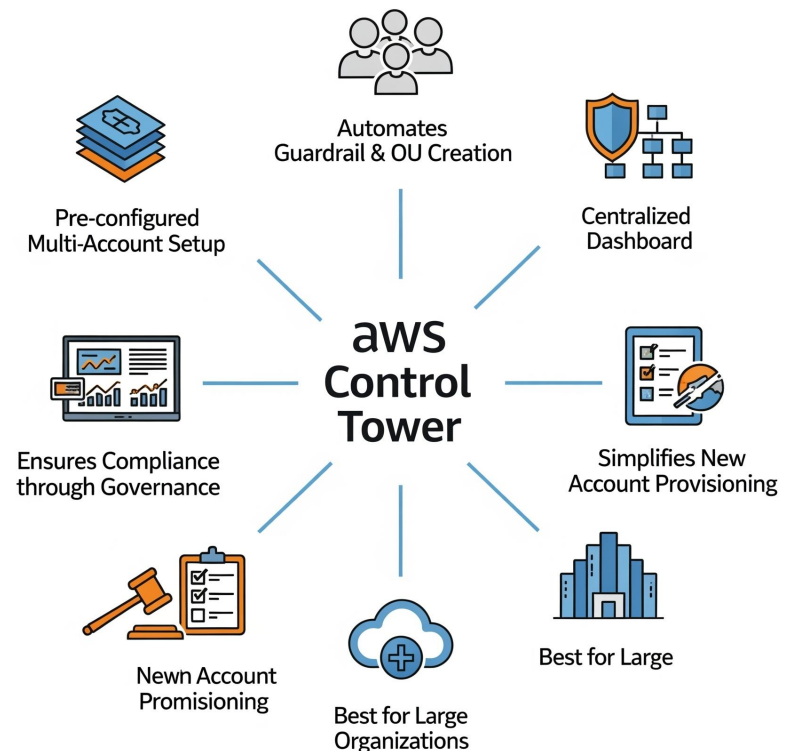
Cost Allocation Strategies

- Use tagging for cost visibility
- Enable Cost Explorer for reports
- Configure AWS Budgets per OU
- Integrate with AWS CUR for analytics
- Forecast usage across departments
- Encourage cost ownership



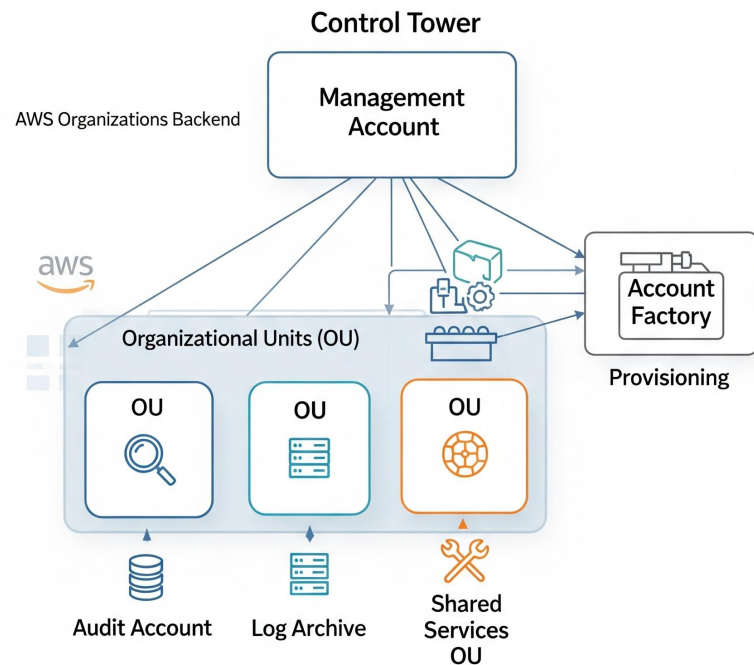
Introduction to AWS Control Tower

- Pre-configured multi-account setup
- Automates guardrail and OU creation
- Provides centralized dashboard
- Ensures compliance through governance
- Simplifies new account provisioning
- Best for large organizations



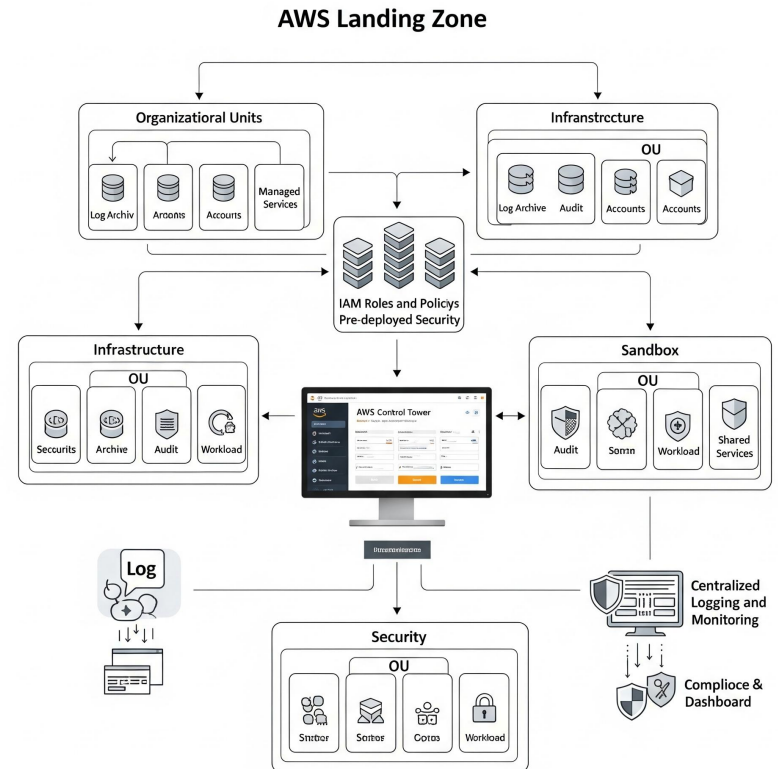
Control Tower Architecture

- Management account controls all OUs
- Audit account for compliance checks
- Log archive for centralized logs
- Shared services OU for IT tools
- Account Factory for provisioning
- Uses AWS Organizations backend



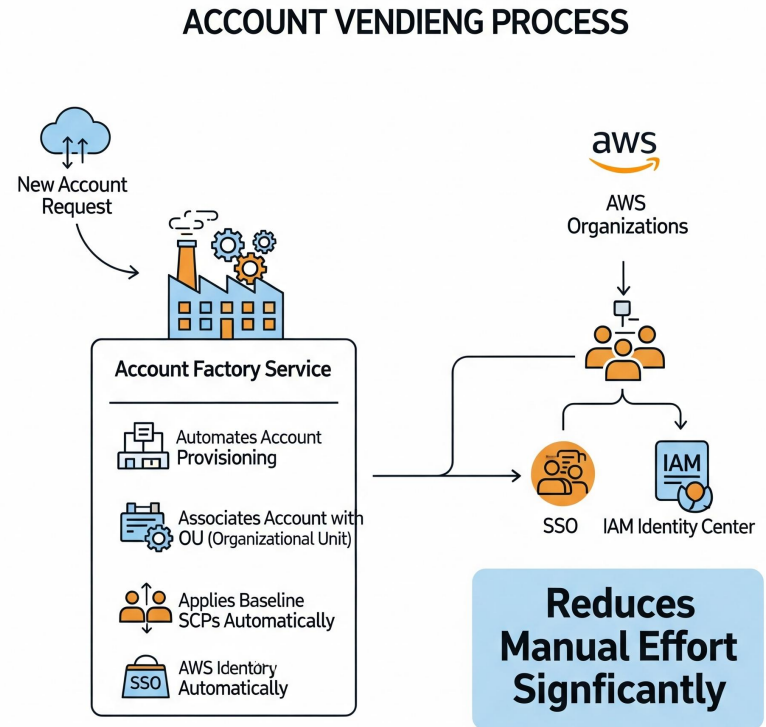
Landing Zones

- Standardized AWS environment baseline
- Includes foundational OUs and accounts
- Pre-deployed IAM roles and policies
- Enables central logging and monitoring
- Supports security & compliance needs
- Managed via Control Tower console



Account Vending Process

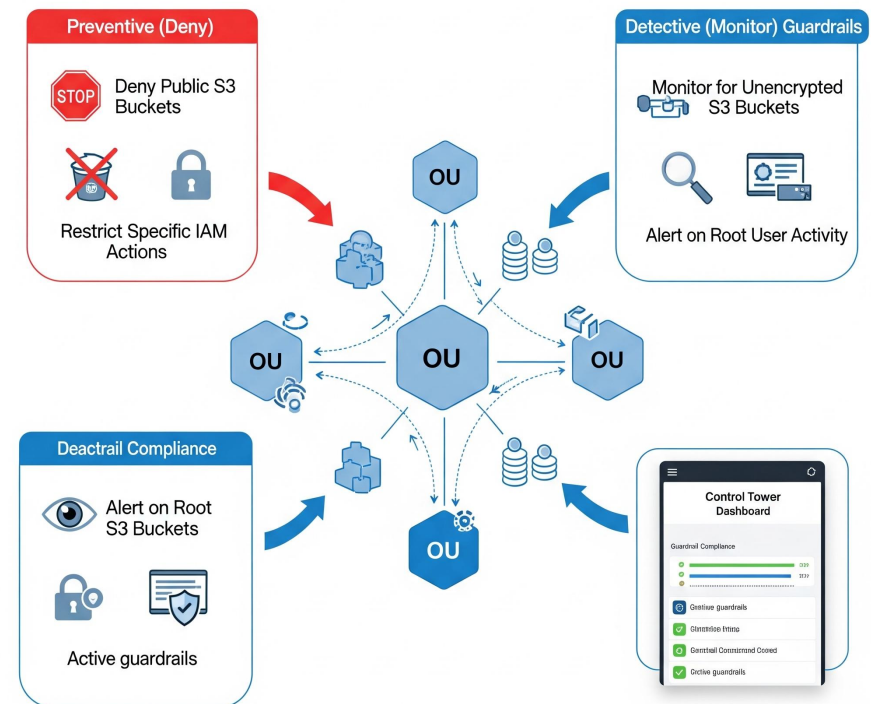
- Uses Account Factory service
- Automates account provisioning
- Associates account with OU
- Applies baseline SCPs automatically
- Integrates with SSO & IAM Identity Center
- Reduces manual effort significantly



Guardrails in Control Tower

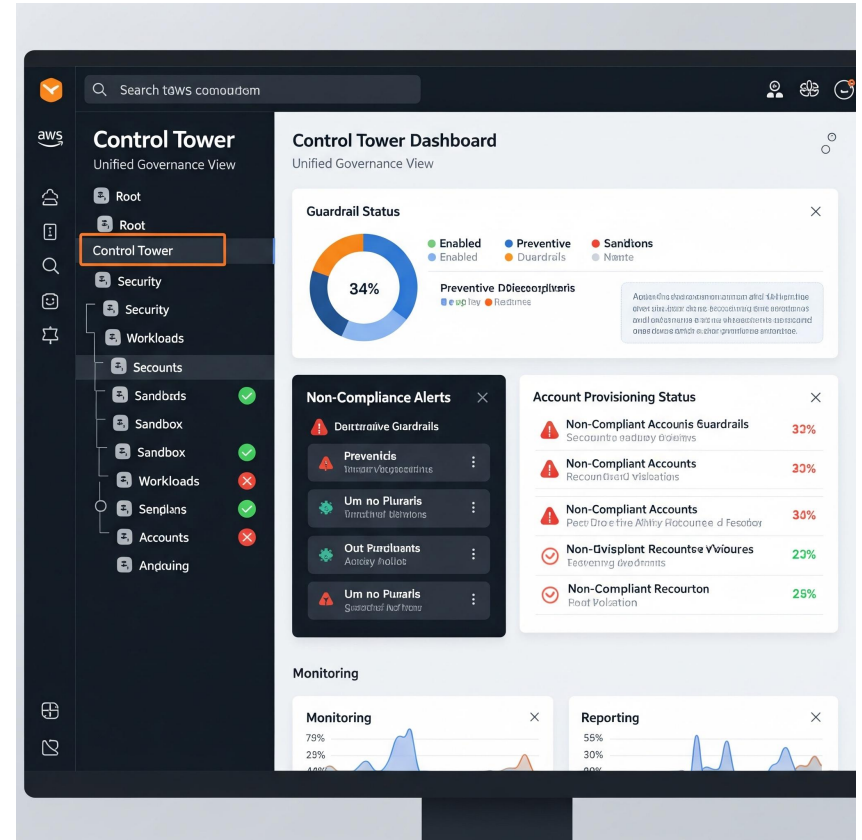
- Pre-defined governance rules
- Preventive (deny) & detective (monitor) types
- Applied automatically to OUs
- Ensures compliance with org policies
- Examples: No public S3 buckets
- Visible in Control Tower Dashboard

Guardrails in Control Tower



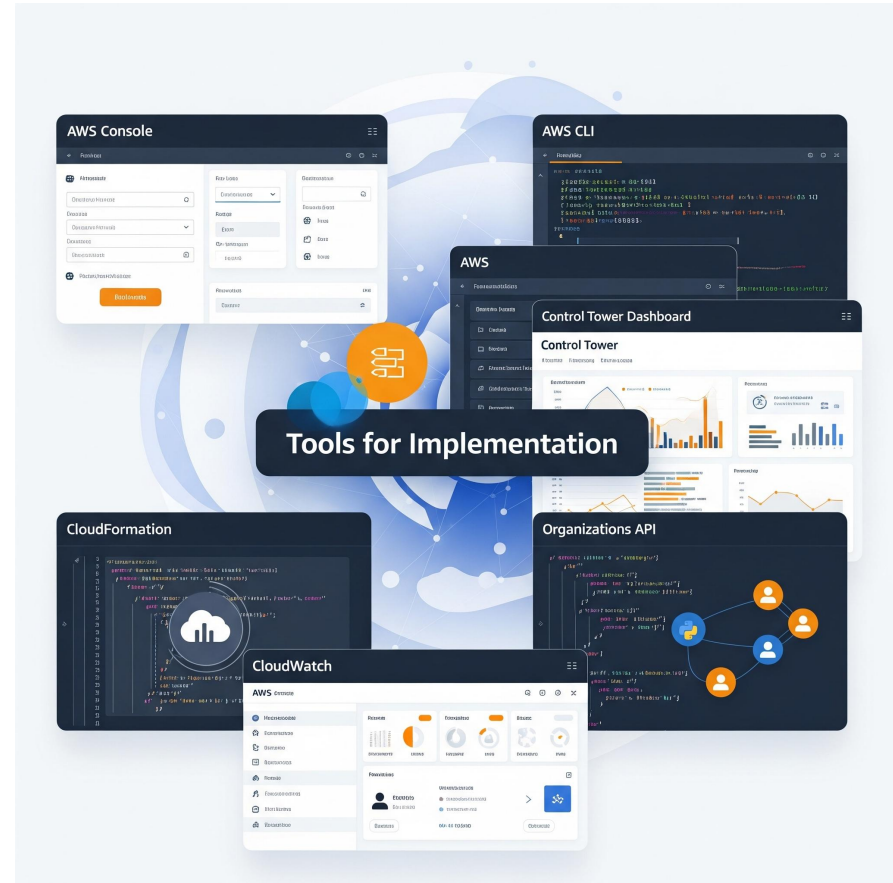
Control Tower Dashboard

- Unified governance view
- Shows OU hierarchy & guardrail status
- Displays non-compliance alerts
- Tracks account provisioning status
- Simplifies monitoring and reporting
- Accessible via AWS Console



Tools for Implementation

- AWS Console: GUI-based setup
- AWS CLI: Command-line automation
- Control Tower Dashboard: Visual governance
- CloudFormation: Infrastructure as Code
- Organizations API: Custom scripting
- CloudWatch: Monitoring and alerts



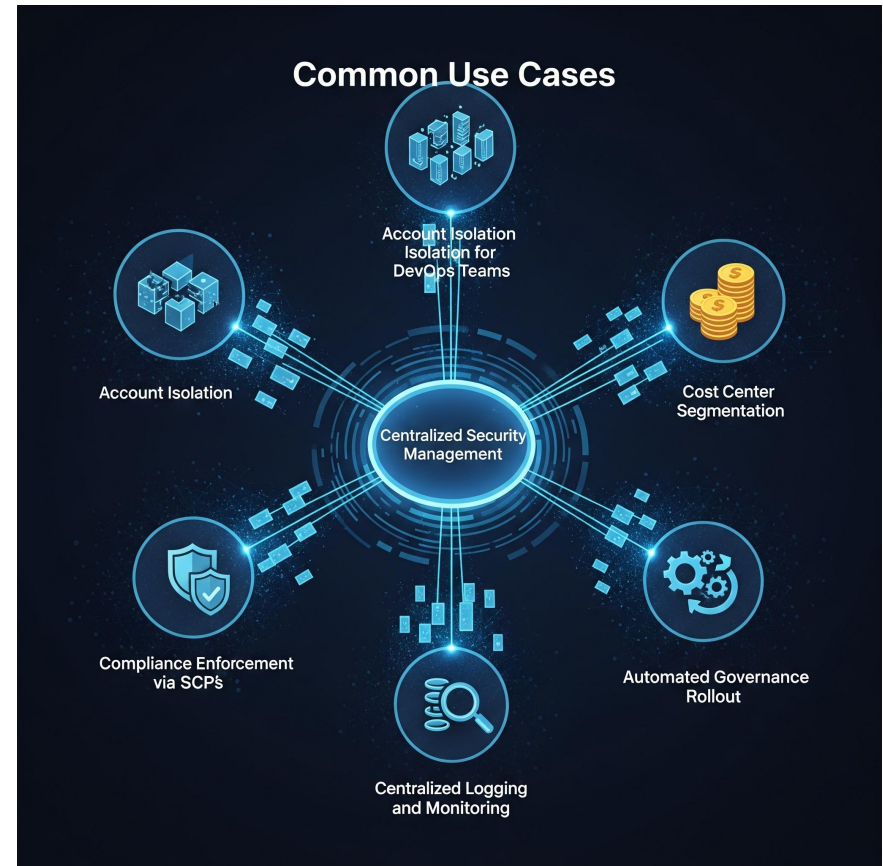
CLI Commands for Organizations

- aws organizations create-organization
- aws organizations create-account
- aws organizations attach-policy
- aws organizations describe-account
- aws organizations list-accounts
- Enables automation pipelines



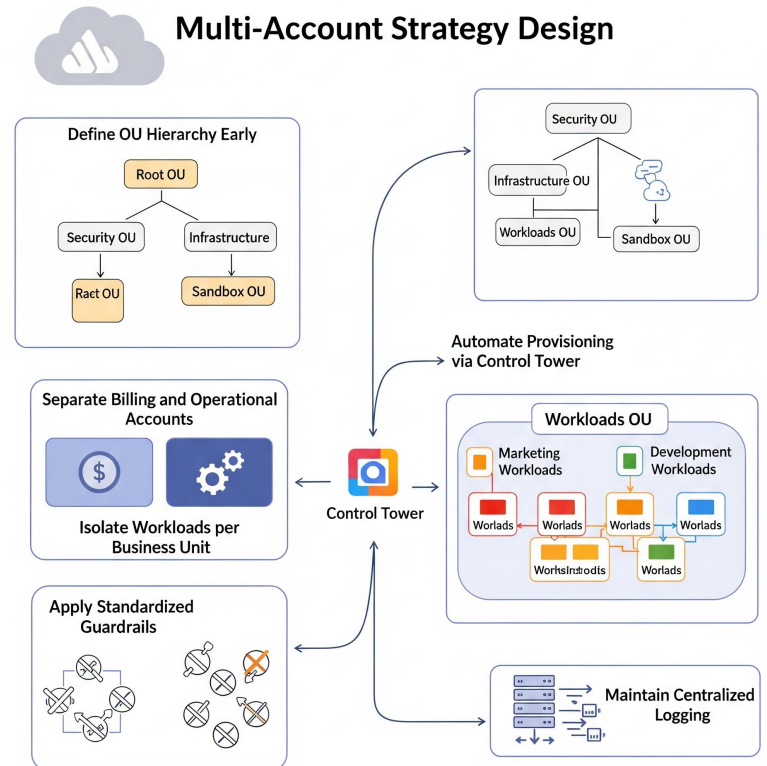
Common Use Cases

- Centralized Security Management
- Account isolation for DevOps teams
- Cost center segmentation
- Compliance enforcement via SCPs
- Centralized logging and monitoring
- Automated governance rollout



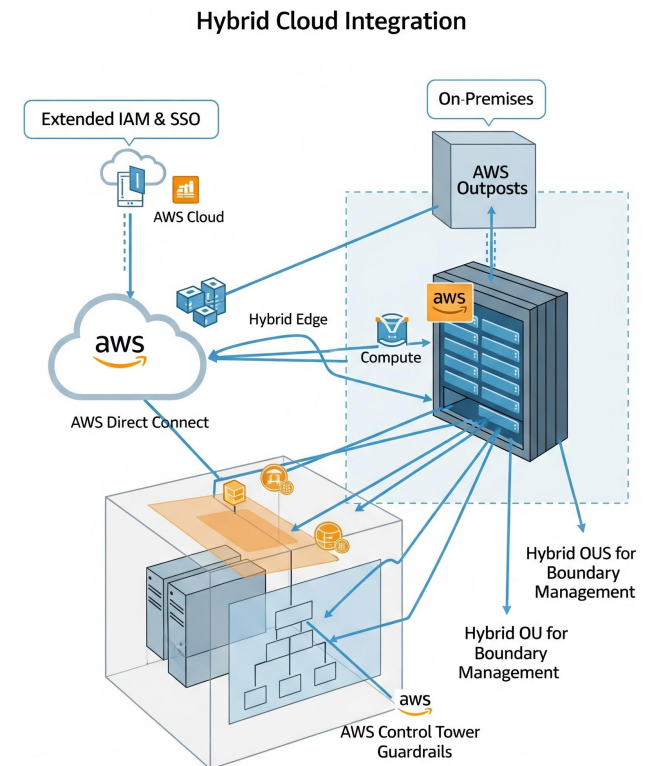
Multi-Account Strategy Design

- Define OU hierarchy early
- Separate billing and operational accounts
- Isolate workloads per business unit
- Automate provisioning via Control Tower
- Apply standardized guardrails
- Maintain centralized logging



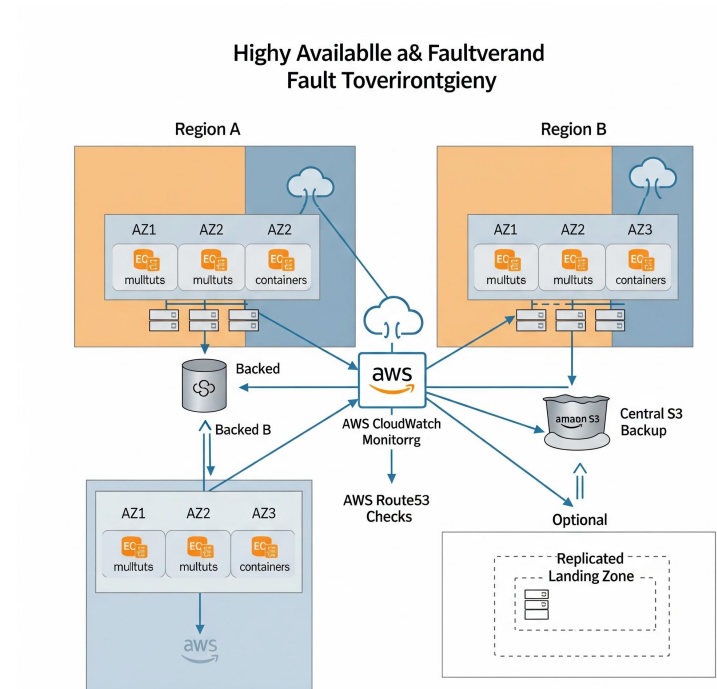
Hybrid Cloud Integration

- Connect AWS with on-prem via Direct Connect
- Use AWS Outposts for local compute
- Enable seamless hybrid workloads
- Extend IAM & SSO to hybrid edge
- Hybrid OUs for boundary management
- Combine with AWS Control Tower guardrails



High Availability & Fault Tolerance

- Deploy workloads across multiple regions
- Implement multi-AZ architecture
- Backup critical data to central S3
- Monitor using AWS CloudWatch
- Utilize AWS Route53 health checks
- Replicate landing zones if needed



Hands-On Implementation Roadmap

- Create AWS Organization and OUs
- Define and attach SCPs
- Set up consolidated billing
- Launch Control Tower landing zone
- Configure guardrails and auditing
- Validate with test accounts

Summary & Key Takeaways

- AWS Organizations simplify governance
- Control Tower automates landing zone setup
- SCPs enforce compliance and restrictions
- Multi-account design boosts scalability
- Guardrails ensure continuous compliance
- Unified tools provide enterprise visibility